NAVIGATING THE FACTOR ZOO

The Science of Quantitative Investing

Michael Zhang, Tao Lu
and Chuan Shi

ROUTLEDGE

NAVIGATING THE FACTOR ZOO

Bridging the gap between theoretical asset pricing and industry practices in factors and factor investing, Zhang et al. provides a comprehensive treatment of factors, along with industry insights on practical factor development.

Chapters cover a wide array of topics, including the foundations of quantamentals, the intricacies of market beta, the significance of statistical moments, the principles of technical analysis, and the impact of market microstructure and liquidity on trading. Furthermore, it delves into the complexities of tail risk and behavioral finance, revealing how psychological factors affect market dynamics. The discussion extends to the sophisticated use of option trading data for predictive insights and the critical differentiation between outcome uncertainty and distribution uncertainty in financial decision-making. A standout feature of the book is its examination of machine learning's role in factor investing, detailing how it transforms data preprocessing, factor discovery, and model construction. Overall, this book provides a holistic view of contemporary financial markets, highlighting the challenges and opportunities in harnessing alternative data and machine learning to develop robust investment strategies.

This book would appeal to investment management professionals and trainees. It will also be of use to graduate and upper undergraduate students in quantitative finance, factor investing, asset management and/or trading.

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PREFACE

> "It is difficult to make predictions, especially about the future."
>
> *(Danish physicist and Nobel laureate Niels Bohr)*

Generations of financial researchers and practitioners tried to predict the financial market. Few of them succeeded.

Eugene Fama, an American economist known for his work on the Efficient Market Hypothesis (EMH), wrote several papers that discuss the unpredictable nature of the market. Two of his most influential papers are:

- Fama, Eugene F. (1965). "The Behavior of Stock-Market Prices". *Journal of Business*, 38(1): 34–105. In this paper, Fama suggests that stock prices follow a random walk, making them unpredictable based on past price information. This laid the foundation for the development of the EMH.
- Fama, Eugene F. (1970). "Efficient Capital Markets: A Review of Theory and Empirical Work". *Journal of Finance*, 25(2): 383–417. In this seminal paper, Fama provides a comprehensive review of the existing theory and empirical work on market efficiency up to that time. He argues that the market is efficient in reflecting all available information, making it difficult for investors to consistently outperform the market on a risk-adjusted basis.

These papers, among others, form the basis of Fama's influential work on the EMH, which suggests that the market is generally unpredictable because of its efficiency in incorporating available information into asset prices.

At the same time, Warren Buffett, the renowned investor and CEO of Berkshire Hathaway, has been quoted as saying, "I'd be a bum on the street with a tin cup if the markets were always efficient." The statement reflects Buffett's skepticism about the EMH.

Indeed, the market cannot be efficient if some investors, such as Warren Buffett and Jim Simons, can consistently beat the market over the course of some 30 years. The question is: how?

This book delves into the hidden ingredients that enable quantitative trading to outperform the market. Ironically, these hidden ingredients, or "factors", were developed by Eugene Fama and his fellow financial economists, who themselves do not believe that the market can be consistently outperformed.

The term "factors" is ubiquitous in finance and asset management. It has been a major focus in asset pricing literature and an active research area within the industry for the past 50 years. However, confusion usually comes when we read academic journals where the term "factors" seems to have a very different meaning from what industry practitioners refer to. From academia's point of view, factors are *market-wide*, which explains the cross-sectional variation of assets' returns, and assets have different returns because of their different *exposure* to the factors. The classical yet famous single-factor model is the Capital Asset Pricing Model (CAPM), which states that the market excess return is the single factor and assets' returns are determined by their systematic exposure (beta) to the market. In the multi-factor space, the Fama-French three-factor model (FF3) starts the race, complementing the market factor with HML (high-minus-low) and SMB (small-minus-big), which are respectively the spread between stock returns of high book-to-market value and low book-to-market value, and the spread between the returns of small market capitalization stocks versus big market capitalization stocks. Following the literature, Carhart (1997) four-factor model complements the FF3 with an additional momentum factor (MOM). Since then, market excess return, HML, SMB, and MOM have become consensus factors within the academic world. Later developments are the liquidity factor (Pástor and Stambaugh 2003) and the tail risk factor (Almeida et al. 2017) etc. In parallel, Fama-French came up with another five-factor model by complementing the FF3 with two additional factors called RMW (robust-minus-weak), which is the spread of returns of firms with robust vs weak operational profitability, and CMA (conservative-minus-aggressive), which is the spread of returns of firms that invest conservatively vs those that invest aggressively.

In contrast, the terminology in the industry is much less rigorous and loosely defined. In this context, they seem to be referring to *variables* or *attributes* of assets that are associated with their future returns. Unlike the academic factors, factors in the industry are not considered market-wide measures. For example, the volatility of stocks correlates with stocks' next-period returns, so volatility could be treated as a factor. While both academia and industry try to use factors to capture *cross-sectional variation*, which essentially tries to answer the question why different assets have different returns during a certain period, the definitions, motivations, treatments, and uses of factors are vastly different, leading to confusions about what exactly factors are, their uses and limitations, etc.

Structure of this Book

We aim to offer a map of factor investing in this book. Each chapter covers an important direction of factor development. Chapter 1 starts with the current landscape in factor investing. It contrasts the definitions, treatments, and identifications of factors between academia and industry, with a phenomenon on the numerous factors being used known as the "factor zoo". Chapter 2 introduces fundamental analysis and a new trend of combining fundamentals with quantitative analysis, known as the "Quantamentals". Chapter 3 studies how various statistical moments relate to risk and are treated as factors. Chapter 4 deals with a crucial risk concept, systematic risk, or beta, as a factor. Chapter 5 introduces factors from price history and technical indicators, with a special focus on trend and jump analysis. Chapter 6 examines microstructure and a broad range of liquidity metrics, from early studies on volume impacts to contemporary advancements in trade time metrics and liquidity resilience. Chapter 7 discusses tail risk, a topic gaining increasing traction. Chapter 8 focuses on factors from behavioral finance, which is a very broad and hot topic on its own. In Chapter 9, we examine how other asset classes like commodities and options may affect equities' returns. Chapter 10 focuses on recent advances in our understanding of distribution uncertainty. Chapter 11 discusses factor development based on alternative data. Finally, in Chapter 12, we discuss various ways of how machine learning may contribute to factor investing. The Epilogue concludes our journey through the factor zoo and offers our reflection on the multifaceted world of factor investing.

The Philosophy of Scientific Investing

Two authors of this book, Zhang and Lu, come with both academic backgrounds—as business school professors—and industry experience, having co-founded a quantitative hedge fund named Super Quantum Fund. Shi manages a hedge fund called Beijing Liangxin Investment Management. This book is written with practitioners in mind, including those from private or public equity funds, asset managers, brokers, financial advisors, and regulators. We hope it serves as a practical guide to factor investing from a scientific perspective, linking flourishing academic research with the industry's practical implementations. The book aims to provide practitioners with a synthesis and formal treatment of factors used in the industry. It also explains how these factors relate to academic foundations with sound economic rationale.

While many market participants in the quantitative trading space emphasize the value of (1) identifying more factors, and (2) developing more complex models, we hold the view that the science behind these factors and models is even more important. Based on our research in academia and practice in the financial market through the Super Quantum Fund, we developed a framework for students and asset managers to gain a deeper understanding of factor investing. In this book, we show how such an understanding can bring consistent returns that transcend both bull and bear markets.

The pyramid's apex comprises *Finance*, encompassing risk management, trading execution, compliance, and operations. Every financial institution should possess this layer to ensure the proper execution of investment strategies to serve investors.

The second layer, *Technology*, relies on inductive reasoning and should be present in all quantitative funds. It incorporates financial engineering aspects such as factor creation, machine learning, and portfolio optimization.

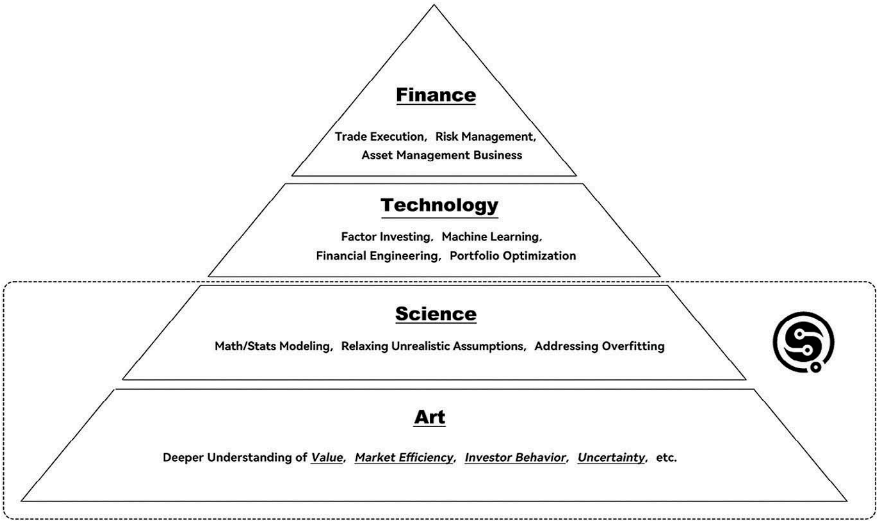

FIGURE 0.1 The Super Quantum Pyramid of Quantitative Investment
Figure shows the Super Quantum pyramid of quantitative investment. It has four layers: Finance, Technology, Science, and Art.

Following this is the *Science* layer, emphasizing the significance of deductive reasoning. It involves creating theoretical and methodological tools to tackle issues unaddressed in the *Technology* layer. For instance, data-driven decision-making is sometimes criticized for its inability to foresee future outcomes: "Predicting future events based on historical occurrences is akin to driving while only looking in the rear-view mirror". To circumvent this limitation, the *Science* layer must develop a solid theory before validating it with data.

The base of the pyramid is *Art*, requiring hypotheses formulated on powerful intuition and an in-depth understanding of financial markets. This layer necessitates a thorough grasp of the inherent uncertainties in the market. As either humans or their devised models constitute market participants, a resilient quantitative strategy must account for trading outcomes that may diverge from logical predictions.

Each layer in the pyramid is crucial, building upon the foundation beneath it to create a holistic framework that fuses intuition, theory development, technological infrastructure, and financial judiciousness to generate superior investment outcomes.

It is clear, from this figure, that financial engineering, machine-learning models, factors, etc. are just a small part of the whole framework. Developing better models should go beyond the Technology layer and push the frontiers of the Science and the Art layers.

GitHub Repository

While writing this book, we developed an open-source code project: the Factor Investing Research Engine (FIRE). We invite our readers to contribute to this project. You can find the project on GitHub at this link: https://github.com/fire-institute/fire. Our goal is for FIRE to progressively encompass all the factor topics discussed within and beyond this book, translating factor research ideas into executable code and ensuring their scientific validity through validations.

While a book remains static once printed, ideas should be dynamic. With FIRE, we can continuously update and expand on new research topics, delve deeper into issues mentioned in the book, and explore fresh insights. The book will remain a living document.

We hope that FIRE serves as a bridge between us, the authors, and you, the readers. We look forward to discussing topics related to factor investing and gaining new inspiration through our interactions. Perhaps these interactions will lead to a series of follow-up works.

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This book is not just a product of our efforts, but a testament to the collective spirit and dedication of all those who have been involved, directly or indirectly.

We are deeply indebted to our spouses and children, who have shown endless patience and understanding during the countless hours we dedicated to this project. Their sacrifices and encouragement have been the bedrock of our perseverance. We also thank our colleagues for their invaluable contributions throughout the journey of writing this book. Their unwavering support and insightful feedback have been pillars of strength and inspiration, without which this work would not have been possible.

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We would like to thank Erik Brynjolfsson, Shane Greenstein, Campbell Harvey, Terrence Hendershott, Yuriy Nevmyvaka, Michael Saunders, and Feng Zhu for their endorsement and valuable suggestions for this book.

We acknowledge the broader community of researchers and professionals in our field. The body of work they have created has been a source of knowledge and motivation, prompting us to contribute to the ongoing dialogue with this book.

To everyone who has played a part in this journey, we offer our heartfelt thanks.

Michael Zhang, Tao Lu, and Chuan Shi

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1

FACTOR INVESTING

In the complex world of financial markets, factors offer a systematic approach to investment by focusing on measurable characteristics that have historically been linked to higher returns. Factors such as value, size, momentum, quality, and volatility allow investors to construct portfolios with a structured methodology, providing a sense of clarity amidst the market's inherent unpredictability.

Quantitative models play a crucial role in factor investing, enabling investors to screen and select securities that exhibit desired factor characteristics. By using historical data and statistical methods, investors can identify which factors are likely to contribute to performance in different market conditions. The increasing use of machine learning and artificial intelligence has further refined these models, allowing for a more nuanced understanding and application of factors.

One of the key benefits of factor investing is the potential for risk management through diversification. By investing in multiple factors, one can mitigate the impact of any single factor underperforming. This approach does not guarantee protection against losses but is intended to achieve a more stable performance over time.

While no investment strategy can offer complete certainty, factor investing provides a disciplined framework that helps mitigate some of the randomness and emotion in investment decisions. By relying on evidence-based factors, investors can make more informed choices with a clearer understanding of the risks and potential returns.

1.1. Background

Factor investing is a strategy in the financial market that involves identifying specific drivers of returns. Factors are quantifiable characteristics of securities that can explain the risk and return profile of an asset or portfolio. The concept is that certain underlying factors or characteristics drive the expected return of an

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investment. They can be understood as *features* in machine learning or *independent variables* or *predictors* in regression analysis.

Factors in investing are fundamental traits or features that are common across a broad set of securities that can help explain their performance and risk profile. These factors are often grounded in academic research and can be economic (e.g., growth, inflation), statistical (e.g., volatility, momentum), or behavioral (e.g., investor sentiment).

Factors are important because they provide a framework for portfolio construction and risk management that is more nuanced than traditional asset-class-based investing. They allow investors to:

- **Target Returns:** By understanding the factors that drive returns, investors can construct portfolios that are more likely to meet their return objectives.
- **Manage Risks:** Factors can expose hidden risks in a portfolio that might not be apparent when looking only at asset classes.
- **Improve Diversification:** Factors can help identify and invest in uncorrelated sources of potential returns, reducing risk through better diversification.
- **Enhance Transparency:** Factor investing offers a clear and transparent method to track the performance and risk profile of an investment strategy.

A "factor" in math means a part that helps make up a whole. For example, the factors of 15 are 3 and 5 because $3 \times 5 = 15$. Similarly, when we "factor" investments, we break down the returns into different parts that contribute to the total performance.

To picture factor investing, think of returns in a multi-dimensional space. Each dimension represents a different factor affecting the returns. By looking at each dimension, we can see how various aspects of the financial world impact an investment's performance.

Dimensions are analogous to the different sources of risk and return that can affect an asset's performance. When analyzing an investment, we are essentially trying to understand its behavior in a space defined by multiple dimensions, each aligned with a specific factor.

Each factor can be thought of as an axis in this multidimensional space. Common factors like size, value, momentum, quality, and volatility act as vectors or directions in this space. Just as in a three-dimensional physical space, where you have axes for length, width, and height, in factor investing, the axes might be these various characteristics of stocks.

When constructing a portfolio, we are projecting our investment onto these various factor dimensions. The projection onto each axis (factor) gives us a sense of the portfolio's sensitivity to that factor. For instance, a high projection on the "value" axis suggests a portfolio heavily weighted towards value stocks.

By analyzing the factor exposures of a portfolio, investors can understand the drivers of performance. If a portfolio outperforms, factor analysis can help determine which factors contributed to this outperformance. For example, if the value factor had strong returns over a period, and the portfolio was heavily projected onto the value dimension, this factor likely played a significant role in the portfolio's performance.

In practice, multi-factor models quantify the contribution of each factor to expected returns using statistical methods. Regression analysis, for instance, can be used to estimate the sensitivity of an asset's returns to each factor, often expressed as beta coefficients. These coefficients measure the extent to which the asset moves in relation to a unit change in the factor.

Traditional multi-factor models in finance are often constructed using linear regression, which assumes that the return of an asset is a linear combination of various factors. However, the assumption that factors are orthogonal to each other (meaning that they are uncorrelated and therefore provide independent information) is an ideal that often does not hold true in practice.

In theory, having orthogonal factors in a multi-factor model is desirable because it simplifies the analysis and interpretation of the model. Orthogonal factors would mean that each factor explains a distinct portion of asset returns, and there is no overlap in the information that each factor provides. This would allow for clear attribution of performance to each factor.

However, in the real world, financial factors often exhibit some degree of correlation. Many factors are empirically correlated to some extent. For example, the value and size factors may be related since smaller companies might also be value companies. Correlation among factors can lead to multicollinearity in regression models, which can inflate the variance of the estimated coefficients and make the model less stable and more sensitive to changes in the model's inputs. To address the issue of non-orthogonality, statistical techniques such as principal component analysis (PCA) can be used to transform correlated factors into a set of uncorrelated factors. These uncorrelated factors are linear combinations of the original factors and are orthogonal by construction. Another technique is factor rotation, which also aims to create factors that are more interpretable and less correlated with each other. This can help in better understanding the unique contribution of each factor.

The advent of machine-learning tools has significantly advanced the field of factor investing by allowing for the discovery and exploitation of complex, non-linear relationships between factors and asset returns. Traditional linear models, such as multiple regression, are limited to linear relationships and assume that the impact of a one-unit change in a factor on asset returns is constant, regardless of the level of the factor. Machine-learning models, on the other hand, can capture more nuanced interactions and non-linear effects.

Machine learning includes a variety of algorithms capable of modeling non-linear relationships. Decision trees, for example, can capture non-linear patterns by splitting the data into branches based on factor thresholds. Ensemble methods like random forests or gradient-boosting machines combine multiple decision trees to improve predictive performance and capture complex factor interactions. At the same time, machine learning can automatically perform feature engineering, which involves creating new features from existing ones to better capture the underlying patterns in the data. For instance, it could involve transforming factors through polynomial expansion to capture non-linear effects or interaction terms to capture the combined effect of multiple factors. Techniques like principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE) can reduce the dimensionality of the data, helping uncover the underlying structure and relationship between factors that may not be apparent in higher-dimensional space. Machine-learning models, such as Lasso and Ridge regression, include regularization techniques that penalize the complexity of the model. This not only helps in preventing overfitting but also in dealing with multicollinearity among factors by shrinking less important factor coefficients toward zero.

Although machine-learning models, especially the more complex ones, are often considered "black boxes", there is a growing field of model interpretability aimed at understanding the decisions made by these models. Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) can help quantify the contribution of each factor to the predictions made by a machine-learning model.

In summary, factor investing models help dissect the complex financial returns into more understandable components. By examining each dimension separately, investors can make more informed decisions about where to allocate their capital to optimize their risk-adjusted returns.

The rest of this chapter is organized as follows. Section 1.2 discusses the role of quants, or quantitative analysts, who are central to developing factor models in finance. Section 1.3 introduces the motivation for understanding factors. Section 1.4 discusses the current landscape of factor investing. In Section 1.5, we compare the definitions and treatments of factors between academia and industry practitioners. Section 1.6 explains the process of identifying factors. Section 1.7 discusses the consensus factors in the asset pricing literature. Section 1.8 provides a broad classification of factors being used in the industry. Section 1.9 discusses factors from the perspective of them as explainers or predictors. Finally, Section 1.10 concludes with a summary table that compares key differences.

1.2. Quant

Factors and factor models are often developed by quantitative analysts, who are commonly referred to as quants. The word "quant" is probably one of the most frequently used prefixes in numerous finance programs offered by universities or job titles from various financial institutions in the past 20 years. Adding the word quant in front of an academic program or a job role seems to give some kind of prestige, at least to the applicants seeking a career in the finance field. From an educational perspective, we observe innovative combinations of terms such as *quantitative, mathematical, computational*, and *engineering* integrated into novel program names like Quantitative Finance, Financial Engineering, Computational Finance, and Mathematical Finance. From a job role perspective, we see quantitative analysts, quant researchers, quantitative portfolio managers, financial engineers, quantitative developers, etc. Furthermore, this quant prefix seems to appear everywhere in banks, and financial institutes like insurance companies, asset management firms, and hedge funds, covering different parts of the firms like front office quants, risk quants, quant developers, etc. What is quant? What are the so-called quant skills? What are the quant skills that are taught by most programs from universities? What are the quant skills required by different roles like banks or hedge funds? The answer lies in the P Quant vs Q Quant distinctions.

Whenever they hear the word quant, most people will assume it refers to investment banks' (like Goldman Sachs or Morgan Stanley) quantitative research that deals with derivatives and structured products pricing if the role sits in the front office. This is unsurprising since this was the focus of the traditional finance programs offered by most universities with the prefix quant or other similar words we mentioned. The origin traces back to the emergence of derivatives and structured products popular since the late 1990s and early 2000s which gave rise to demand for heavy mathematical tools that were previously only used in science and engineering. The quant in this respect is usually referred to as the Q Quant which generally covers quantitative researchers working on derivatives valuation, a function mostly found in the sell side. The term risk-neutral measure is usually represented by the letter Q in the literature. Derivative pricing is based on hedge arguments, which means the perfectly hedged instrument/portfolio is essentially risk-free. With hedges, risk-averse individuals would behave like risk-neutral ones in the Q world. The transformation from the actual world (the P world) to the Q world lies in the *risk premium*, $(r - r_f)$ where r is the expected rate of return of a security in the real world, whereas r_f is the risk-free rate (the rate of return in a hedged, or risk-free world). The concept of risk premium is a core idea in quantitative finance, and it has key implications in both the P and Q worlds. Q Quants aim to price a security (mainly derivatives) fairly. The pricing requires the calibration of stochastic processes based on the underlying stocks' prices and other information such as dividends etc. The main idea is to *extrapolate the present* from available information.

Alternatively, P Quants refer to quantitative investors from the buy side such as quantitative hedge funds (e.g., Citadel, Two Sigma, AQR Capital, etc.) that focus on constructing quantitative portfolios by some kind of combination of statistics and machine-learning techniques for trading purposes. The use of P comes from an *objective* probability measure to model possible states in the future. In this spectrum, multivariate statistics and advanced econometrics are used for estimation to determine what securities to purchase and to sell to get the highest chance of a positive profit-and-loss profile. This results in rigorous portfolio construction, trading strategy, and risk management. In short, it is to model the future based on current information for profit-making purpose.

Quantitative factor investing belongs to the P Quant world since we want to find factors that could predict positive returns and trade on them. Nevertheless, the Q world might complement in some way. This is because some factors might need techniques from Q world for fair pricing, especially those related to derivatives. We will see an example of the jump factor which applies variance swap pricing technique from the Q world in Chapter 9.

Although the P world and Q world seem to be very different, there are occasions when the two worlds intersect. One example is derivatives hedging where fund managers in the P world need to use the "Greeks" calculated from the Q world to manage their trading book's risk. Another example can be given in the context of statistical arbitrage, where arbitragers use the fair prices of derivatives from the Q world as equilibrium and bet on real-price deviations from the equilibrium in the hope of price reversion.

1.3. Factor Investing and Value Investing

Factor investing and value investing are two distinct approaches within the investment world. Factor investing is a broad strategy that aims to utilize multiple factors—quantifiable characteristics of securities that can explain differences in returns—to construct and manage a portfolio. It is typically systematic and relies on a rules-based approach to identify and capitalize on these factors, such as size, value, momentum, quality, and low volatility, among others.

Value investing was made well-known and significant by Benjamin Graham and David Dodd, often referred to as the "fathers of value investing". Their seminal work, *Security Analysis*, first published in 1934, laid the intellectual foundation for this investment philosophy. Benjamin Graham later elaborated on these concepts in his book *The Intelligent Investor*, which was first published in 1949 and has since been widely regarded as one of the most important books on investing. Warren Buffett, perhaps the most famous proponent of value investing, was a student of Benjamin Graham at Columbia Business School and worked for Graham before starting his own investment partnership. Buffett's success at Berkshire Hathaway has made him an iconic figure in the investment world and has further popularized the value-investing approach. His annual letters to shareholders of Berkshire Hathaway are read extensively for insights into the practical application of value-investing principles.

Value investing focuses on finding securities that appear undervalued compared to their intrinsic value, often assessed through fundamental analysis. Value investing is epitomized by the search for stocks trading at a discount to their book value or with low price-to-earnings ratios, and it frequently involves a long-term horizon, with investors willing to wait for the market to realize the true value of these assets.

Despite their distinct characteristics, factor investing and value investing can exhibit similarities in certain aspects: Both strategies are grounded in the idea of rational investment, meaning they both rely on the analysis of financial data to make investment decisions. They reject speculative approaches that are based on hype without underlying financial justification. Both styles often use financial metrics to select investments. While factor investing may use a broader set of factors, value investing specifically focuses on value indicators such as the Price-To-Earnings (P/E) ratio, Price-To-Book (P/B) ratio, and dividend yield. The value factor is indeed one of the key factors in factor investing, illustrating a direct overlap. Both approaches typically take a long-term view of investing. Factor investors believe that certain factors will deliver superior returns over the long term, while value investors wait for the market to recognize and correct the undervaluation of the assets they have purchased. Both strategies aim to manage risk, though they may do so differently. Factor investing manages risk through diversification across different factors, while value investing often considers a margin of safety when purchasing undervalued assets, which can provide a buffer against errors in valuation or unforeseen market downturns. At times, both factor and value investors may take contrarian positions. For example, buying an undervalued stock in a downtrend is a contrarian move typical of value investing. Similarly, a factor investor might invest in stocks with high exposure to the value factor when growth stocks are in favor, which is also a contrarian stance. While factor investing is known for its quantitative approach, modern value investing has also incorporated more quantitative methods, blurring the lines between the two. As a result, many value investors now use quantitative tools to screen for undervalued stocks or to assess risk. Most importantly, both factor and value investors operate under the premise that markets can be inefficient in the short term, which can create opportunities for superior returns. Both styles seek to exploit inefficiencies such as mispricing and irrational behavior to generate long-term returns.

In essence, factor and value investing are not mutually exclusive and can sometimes be integrated into a comprehensive investment strategy. While factor investing often employs quantitative models, possibly incorporating machine learning to discern complex relationships, value investing may combine quantitative metrics with qualitative assessments of a company's financial health. Both strategies aim to outperform the market, but they do so through different lenses—one through diversified exposure to proven risk factors and the other through the identification of undervalued assets poised for appreciation.

Table 1.1 outlines some key differences between factor investing and value investing.

We can examine how factor investing works by examining and explaining Warren Buffett's Berkshire Hathaway performance by six factors (see Table 1.2) (Frazzini, Kabiller, and Pedersen 2018). The six factors are respectively excess market return, size factor, value factor, momentum factor, betting against beta factor, and quality factor.

From April 1980 to March 2017, as can be seen in the center panel, if a simple one-factor model (MKT, the market factor only) is applied to explain Berkshire's portfolio excess return, the alpha (intercept) of the model is 5.8% with a *t*-statistic of 3.09 (an absolute value of *t*-statistics above 2.0 can be considered statistically significant). If a four-factor model is used, with the market factor (MKT), the size factor (SMB), the value factor (HML) and the momentum factor (UMD) combined, the resulting alpha is still statistically significant at a *t*-statistic of 2.46. Suppose a five-factor model is used, that is, adding the Betting-Against-Beta (BAB), the alpha is further reduced, and the *t*-statistic falls to 1.62. If all the six factors are included in the model, the alpha becomes 0.3% with a *t*-statistic of 0.16, which is no longer significant. It is found that the two main factors that make the alpha insignificant are the BAB and the quality factor (QMJ). These six factors explain away Warren Buffet's alpha and at the same time can explain 61% of the variations of the model, suggesting significant explanatory power.

TABLE 1.1 Factor Investing vs Value Investing

	Factor Investing	*Value Investing*
Definition	Investing strategy that targets specific drivers of returns across asset classes.	Investing strategy that involves picking stocks that appear to be trading for less than their intrinsic value.
Investment Focus	Multiple factors (e.g., value, size, momentum, quality, volatility).	Typically focuses on identifying undervalued stocks.
Approach	Systematic, rules-based approach to portfolio construction.	Often involves qualitative analysis and fundamental research to identify undervalued stocks.
Diversification	Seeks to diversify across various factors to reduce risk and enhance returns.	May concentrate on a smaller number of holdings believed to be undervalued.
Quantitative Emphasis	High; typically involves complex statistical models and may use machine-learning techniques.	Moderate to high; relies on quantitative measures of value, but may also include qualitative assessment.
Time Horizon	Can vary; some factors may work better over different time horizons.	Long term; value investors may hold stocks for several years until they reach their perceived intrinsic value.
Market Conditions	Designed to perform across various market conditions depending on the factor exposure.	May perform better in bear markets when undervalued stocks can become more pronounced, or when the market corrects and value stocks may be favored.
Investor Psychology	Less driven by emotions.	Often contrarian, going against prevailing market sentiments to buy undervalued stocks.
Performance Attribution	Returns are attributed to factor premiums and their respective weights within the portfolio.	Returns are attributed to the realization of the asset's intrinsic value over time.

According to the factor model decomposition, Buffett's success can be attributed to the six factors. Moreover, the consistency in exposure to high-value, high-quality and low-beta stocks highlights his emphasis on value investing.

1.4. Current Landscape in Factor Investing

1.4.1. Smart Beta vs Factor Investing

One popular term we encounter in the industry is smart beta strategy. It is worth understanding what this term means and how it comes about.

The term "smart beta" originally referred to alternative index construction rules as opposed to market-capitalization weighted indexes. Beta refers to market exposure that originates from the Capital Asset Pricing Model (CAPM) which states that the expected excess return of any security is determined by its systematic (beta) exposure to the market-risk premium (i.e., the expected market return over the

TABLE 1.2. Berkshire Hathaway Performance Analysis

	Berkshire Stock, 10/1976–3/2017				*13F Portfolio, 4/1980–3/2017*				*Private Holdings, 4/1980–3/2017*			
Alpha	13.4%	11.0%	8.5%	5.4%	5.8%	4.5%	3.0%	0.3%	7.0%	4.9%	3.9%	3.5%
	(4.01)	(3.30)	(2.55)	(1.55)	(3.09)	(2.46)	(1.62)	(0.16)	(1.98)	(1.40)	(1.10)	(0.91)
MKT	0.69	0.83	0.83	0.95	0.77	0.85	0.86	0.95	0.30	0.39	0.40	0.42
	(11.00)	(12.74)	(12.99)	(12.77)	(22.06)	(23.81)	(24.36)	(23.52)	(4.46)	(5.63)	(5.72)	(5.03)
SMB		−0.29	−0.30	−0.13		−0.19	−0.19	−0.05		−0.26	−0.25	−0.23
		(−3.11)	(−3.19)	(−1.17)		(−3.73)	(−3.79)	(−0.95)		(−2.65)	(−2.56)	(−1.95)
HML		0.47	0.31	0.40		0.28	0.19	0.25		0.28	0.21	0.22
		(4.68)	(2.82)	(3.55)		(5.20)	(3.25)	(4.32)		(2.63)	(1.80)	(1.85)
UMD		0.06	−0.02	−0.05		−0.01	−0.06	−0.09		0.08	0.04	0.04
		(1.00)	(−0.25)	(−0.80)		(−0.36)	(−1.66)	(−2.58)		(1.24)	(0.62)	(0.51)
BAB			0.33	0.27			0.19	0.15			0.15	0.14
			(3.79)	(3.04)			(4.08)	(3.18)			(1.61)	(1.53)
QMJ				0.47				0.37				0.07
				(3.06)				(4.55)				(0.43)
$\bar{R}^2$	0.20	0.25	0.27	0.29	0.52	0.58	0.59	0.61	0.05	0.08	0.08	0.08
Obs	486	486	486	486	444	444	444	444	399	399	399	399

Source Values excerpted from Frazzini, Kabiller, and Pedersen (2018).

risk-free rate). One key result of CAPM is that the optimal portfolio offering the highest reward-to-risk ratio (the so-called Sharpe ratio) is the market-capitalization (market-cap) weighted portfolio. Because of the cost-effectiveness of *passive tracking*, ETFs tracking various market indexes have flourished in the market since then. However, with unrealistically rigorous assumptions of CAPM such as frictionless markets and unlimited long-short restrictions, etc., and abundant empirical evidence pointing to the rejection of the model, deviations from passive tracking strategies start to gain traction. These give rise to the idea of smart beta, which could be interpreted as *smart* or *alternative* exposure to the market in contrast to the pure market-cap weighting strategy.

With smart beta schemes, portfolio weights are optimized according to specified objectives by using alternative weighting methods. Popular examples include risk parity, maximum diversification, minimum variance, and equal-weighting schemes. We introduce them briefly here.

Risk parity is an investment strategy that focuses on allocating capital based on risk, rather than on market capitalization or other traditional methods. The goal is to balance the portfolio so that each asset class contributes equally to the overall risk of the portfolio. This contrasts with a more traditional approach where asset allocation might be based on market value, potentially leading to concentration in certain asset classes and, consequently, a higher risk exposure to those areas.

In the context of smart beta, which refers to investment strategies that deviate from the traditional market capitalization-weighted indexes to seek improved returns or lower risk, risk parity can be a guiding principle. Smart beta strategies use alternative methods to construct their portfolios, often based on factors such as value, size, momentum, dividend yield, and volatility.

Risk parity can be integrated into smart beta strategies by ensuring that the portfolio isn't overly reliant on any single factor (or small set of factors) that could lead to disproportionate risk. For instance, a smart beta fund might be constructed to have exposure to both low-volatility stocks and high-dividend stocks, but through risk parity, the fund would allocate its assets so that each of these factors contributes equally to the portfolio's risk profile. This is done by adjusting the investment weights according to the risk each asset or factor brings to the portfolio.

The rationale behind risk parity in the smart beta context is to create a more robust portfolio that can withstand different market conditions. By not overexposing the portfolio to any single source of risk, it may perform more consistently across various market environments. This is particularly appealing to investors who wish to minimize the impact of market volatility and are looking for a more stable return profile over the long term.

Marginal contribution to risk of an asset within a portfolio is defined as

$$w_i \frac{\partial \sigma_p}{\partial w_i}, \tag{1.1}$$

with w_i being the weight of asset i within a portfolio and σ_p the portfolio volatility (the standard deviation of returns)

$$\sigma_p = \sqrt{\Sigma_i w_i^2 \sigma_i^2 + \Sigma_i \Sigma_{j \neq i} w_i w_j \sigma_{ij}}. \tag{1.2}$$

Since $\Sigma_i w_i \frac{\partial \sigma_p}{\partial w_i} = \frac{1}{\sigma_p} \Sigma_i w_i \left(w_i \sigma_i^2 + \Sigma_{j \neq i} w_j \sigma_{ij} \right) = \sigma_p$, the term $w_i \frac{\partial \sigma_p}{\partial w_i}$ is called the marginal contribution to risk as the sum (over all i's) equals the portfolio risk (volatility). The purpose of the risk parity scheme is to choose the weight w_i (for $i = 1, \ldots, n$) such that the marginal contribution to risk is equalized across assets.

Risk parity in Bridgewater's "All-Weather" investment strategy

Risk parity differs from modern portfolio theory which suggests minimizing variance based on a fixed level of expected return. The risk parity method allows capital allocation on a risk-weighted basis to diversify investment (across asset classes), viewing the risk and return of the entire portfolio as one.

The concept of risk parity originated from Bridgewater Associates, a global investment management firm. The firm created a very famous "All-Weather" investment strategy, which formed the foundation of risk parity. The strategy tries to answer the question "What kind of investment portfolio would perform well across all environments?", and it turns out this can be achieved by *balancing* the risk of assets.

Under this framework, assets (stocks and bonds) are balanced according to environmental conditions, such as inflation and growth periods, etc. By equalizing each asset's contribution to risk in a portfolio, the underperformance of some assets could be mitigated by others to a large extent. The term "risk parity" became popular after the successful implementation of the All-Weather investment strategy. It remains one of the most widely used smart beta schemes in the industry.

Maximum diversification scheme aims to maximize the diversification index (DI) by choosing w_i. DI is defined as

$$DI = \left(\frac{\Sigma_i w_i \sigma_i}{\sqrt{\Sigma_{i,j} w_i w_j \sigma_{ij}}} \right). \tag{1.3}$$

The numerator of DI is the weighted average of volatility while the denominator is the portfolio's volatility. To understand its intuition, we take the example of a two-asset case. Suppose there is a perfect correlation (Pearson's correlation of one) between two assets, i.e., no diversification effect. It can easily be shown that the portfolio standard deviation is the weighted sum of the two assets' standard deviations: $\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_1 \sigma_2 (1)} = w_1 \sigma_1 + w_2 \sigma_2$. In this case, Equation (1.3) will equal one, which is also the lower bound. When we extend the problem to multiple assets, we can see that the numerator of Equation (1.3) is always greater than the denominator. If there is a strong internal diversification within a portfolio, the value of DI will be higher. Therefore, maximizing DI means maximizing the diversification effect within a portfolio.

TOBAM and Amundi: A Story on Maximum Diversification Ratio (MDR)

TOBAM is a French asset management company that is credited with developing the concept of the Maximum Diversification Ratio (MDR). The idea behind MDR is to construct a portfolio that maximizes the diversification across its holdings. TOBAM's philosophy posits that diversification, more than any other component, is key to achieving better risk-adjusted returns. This is backed by the idea that markets are not always efficient and that risk can be reduced more effectively through diversification than through traditional asset allocation strategies.

The Maximum Diversification Ratio is calculated as follows:

$$MDR = \frac{Portfolio\ Volatility}{Weighted\ Average\ Volatility\ of\ Individual\ Securities}$$

where *Portfolio Volatility* is the standard deviation of the portfolio's returns and *Weighted Average Volatility of Individual Securities* is the sum of the products of each security's weight in the portfolio and its volatility.

The numerator represents the risk (volatility) of the entire portfolio, while the denominator represents the sum of risks if each asset was held in isolation, weighted by their respective portfolio weights. The work was published in 2006 in the United States Patent and Trademark Office and later in 2008 in the *Journal of Portfolio Management.*

TOBAM's MDR approach uses quantitative methods to assess the diversification of each asset in the portfolio. It then constructs a portfolio that maximizes this diversification measure.

In 2012, Amundi, the largest asset manager in Europe, entered a strategic partnership with French asset management company TOBAM. Amundi provides the platform and distribution channels that help bring TOBAM's approach to a wider base of investors. Together, they have created and offered investment solutions that aim to maximize the diversification benefits for investors.

The MDR-based products have been met with interest from investors looking for alternative ways to construct their portfolios, especially in times of increased market volatility and correlation between asset classes.

Minimum variance, as its name suggests, tries to minimize a portfolio's variance by choosing an optimal weight vector. One can imagine that under this scheme, most weights will be allocated to lower volatility stocks like utilities. Stocks with lower covariances with others will also have higher weights as well.

The equal-weighting scheme is the easiest. It assigns the same weights to all constituents within a portfolio. Despite its straightforward interpretation, this scheme suffers from a liquidity issue since all stocks (liquid or illiquid) are financed equally. When a portfolio has a high capital base and insufficient stocks, the market impact on some illiquid stocks can be prohibitively expensive.

Note that all schemes mentioned above are based on weighting within a portfolio instead of picking the stocks, which involves much more activeness. Smart beta as an alternative weighting scheme can be regarded as somewhere between passive and active management.

The term smart beta later evolved to refer to enhanced-return strategies that involve using factors to outperform a benchmark index. In factor investing, we choose the factors we would like to be exposed to in each environment. ETFs and custom indices that mimic different factors emerge for this purpose. The famous factor ETFs are, for instance, the iShares issued by Blackrock and custom indices produced by MSCI Barra.

1.4.2. The Factor Zoo

Factor ETFs and custom indices are based on broad concepts like value, size, momentum, volatility, quality, etc. They are sometimes termed "styles" instead of "factors" since each of them is composed of many variables (or attributes) or a combination/formula of variables. For example, value consists of various fundamental variables from financial statements or balance sheets like book-to-price ratio, earnings-to-price ratio, book values, sales, earnings, cashflow, etc., and some extended calculations like free cashflow which involves sales revenue, operating costs, taxes, and operating capital in the calculation. These variables are equivocally included as factors in the industry. With multiple styles and many variables within styles, together with ongoing unrelenting efforts to discover new factors, the rate of factor production becomes so fast and out-of-control that many of them become redundant or replicates of others. Professor John Cochrane, a famous financial economist, describes this phenomenon as the "factor zoo".

"Factor Zoo" by John Cochrane

John Cochrane was formerly a professor of economics and finance at the University of Chicago. He serves full-time as the Rose-Marie and Jack Anderson Senior Fellow at the Hoover Institution at Stanford University. His book, *Asset Pricing,* is the standard textbook for graduate courses on asset pricing in finance. It earned him the TIAA-CREF Institute Paul A. Samuelson Award.

Cochrane was active in the media and contributed to the debate on the financial crisis during 2008. His blog *The Grumpy Economist* humorously commented on the issues which drew criticism from Paul Krugman, who won the

Nobel Memorial Prize in Economic Sciences in 2008, leading Cochrane to write a response on his website. The response was later published in *The Wall Street Journal*.

In his 2011 presidential address to the American Finance Association, Cochrane coined the term *zoo of factors*. This was to describe the uncontrolled explosive growth of factors in the industry in a humorous way, which is how the term *factor zoo* came.

Cochrane is the son-in-law of the famous financial economist Eugene Fama, who won the Nobel Memorial Prize in Economic Sciences in 2013 and is one of the most prominent figures in asset pricing literature.

1.4.3 Multiple Hypothesis Testing and False Discoveries

One cause of the "factor zoo" in finance is the phenomenon of multiple hypothesis testing (MHT) and the resultant false discoveries. MHT refers to the practice of testing multiple null hypotheses using the same dataset. In empirical asset pricing and factor investing, the mining of hundreds or even thousands of factors from the same historical data exemplifies a multiple hypothesis testing problem. When testing several hypotheses simultaneously, it's possible to encounter some false discoveries merely by chance.

Consider the scenario where 100 independent factors (i.e., 100 null hypotheses) are tested simultaneously, and one factor shows a t-statistic of 2.0. In this case, we cannot assert that the factor is significant at the 5% significance level. This is because, even if all 100 null hypotheses are true (meaning their expected excess returns are zero), there's still a 99% chance of encountering a t-statistic greater than 2.0 purely by chance. Hence, using the conventional threshold of 2.0 for the t-statistic to determine significance will inevitably lead to many false discoveries or false rejections. This type of error, where a true null hypothesis is incorrectly rejected, is statistically known as a Type I error.

In recent years, to secure publications in top-tier journals, scholars have indulged in data snooping, excessively chasing low p-values (a practice known as p-hacking). Unfortunately, due to deliberate or unintentional data manipulation, the use of lax statistical methods, misinterpretations of p-values, and neglect of the inherent economic logic of factors, many factors concocted under utilitarian motives fail to hold up after publication, as discussed by McLean and Pontiff (2016). Correspondingly, Smart Beta ETF funds also suffer from the problem of p-hacking (Huang et al. forthcoming).

At the 2017 American Finance Association (AFA) meeting, Professor Campbell Harvey, then AFA president, dissected this phenomenon in his speech "The Scientific Outlook of Financial Economics". Advocating for a scientific approach and integrity, Harvey urged scholars to acknowledge and reevaluate the research culture in empirical asset pricing (Harvey 2017).

Professor Campbell R. Harvey

Professor Campbell R. Harvey is a distinguished academic figure in the realm of finance, particularly celebrated for his expertise in investment finance and international finance at Duke University's Fuqua School of Business. With a PhD from the University of Chicago, Professor Harvey has devoted his career to dissecting the intricacies of financial markets and instruments. His scholarly pursuits have firmly established him as a leader in the field, fostering a deeper understanding of market dynamics through his teaching and prolific research.

In the sphere of factor investing, Professor Harvey's contributions are particularly noteworthy. Harvey has delved into the temporal aspects of factor premiums, shedding light on their performance across various economic conditions. This exploration has armed investors with valuable insights about the timing and reasons behind the outperformance of certain factors. Moreover, he has been at the forefront of critiquing the rapid expansion of factors within academic circles. Harvey has argued for more rigorous statistical methods to curb data mining practices and to prevent the recognition of spurious factors that falter outside of sample data.

His innovative methodologies for assessing financial factors have underscored the necessity of solid statistical techniques to circumvent false discoveries in finance. This has been especially significant in a world where investment strategies are increasingly data-driven. Harvey's influence extends beyond academia into the realm of practical finance, where he has educated numerous investment professionals on the nuances of factor investing. His ability to distill complex research into practical investment advice has altered the landscape of factor investing.

Moreover, Campbell Harvey and his long-time collaborator Yan Liu have made significant contributions to this research agenda, which encompasses frequentist and Bayesian methods. In their seminal paper "… and the Cross-Section of Expected Returns" (Harvey, Liu, and Zhu 2016), they mainly flagged the issue, arguing that multiple testing may contribute to the proliferation of factors or asset pricing anomalies. In their subsequent work "Lucky Factors" (Harvey and Liu 2021a), they proposed a general testing framework that nests most asset pricing environments and systematically addresses the issue of multiple testing. In an attempt to generalize the Type I and Type II error trade off to multiple testing, they took their insights to the prominent yet perplexing performance evaluation literature and provided a different answer to the question: "how many fund managers outperform?" (see Harvey and Liu 2020, 2022). Finally, borrowing the idea of prior anchoring from the Bayesian literature on multiple testing adjustments, they proposed a new frequentist approach to forming robust estimates in a panel data setup, as exemplified in Harvey and Liu (2018, 2019).

Addressing MHT and false discoveries has long been a focus in other disciplines, as noted by Ioannidis (2005), while finance has been relatively late to the game.

However, the good news is that both the academic and professional worlds are now recognizing this issue. Regarding the severity of this issue, it is crucial for the academic community to discuss it with an open mind. For instance, Chen (2021) and Jensen, Kelly, and Pedersen (2023) have pointed out that the problem may not be as severe as previously thought. Nevertheless, as Harvey and Liu (2021b) attest, the issue remains under-identified since we only observe published factors, unaware of the total number attempted. As a result, acknowledging this methodological issue and deriving convincing conclusions through sound priors is the right approach to research.

1.5. Factors? What Are They?

We have so far seen the loose definition of factors from an industry point of view, i.e., any variables that seem to drive (or correlate to) future returns on assets. Nevertheless, when we read asset-pricing literature from academic journals, the term "factors" seems to have a completely different meaning. A related and omnipresent term "alpha" also refers to different things between industry and academia. To reconcile the differences, it is worth a brief review of factor models in the asset-pricing literature.

The linear factor model representation of assets' expected returns roots deeply in the theory of asset pricing, which shows the equivalence between the stochastic discount factor (SDF) representation, the mean-variance efficient portfolio, and the linear factor model in terms of pricing assets' returns in the cross-section. While a full treatment of all the necessary mathematical derivation is beyond the scope of this book, a brief introduction about its background is essential to explain why identifying factors and specifying the linear factor model are of paramount importance for both academia and the asset management industry to analyze assets' returns. For a comprehensive discussion of the asset pricing theory, interested readers are referred to the authoritative book of Cochrane (2005).

In the spirit of Ross's (1976) arbitrage pricing theory (APT), the data-generating process of assets' excess returns is governed by

$$r = \mathrm{E}[r] + \beta\nu + e \tag{1.4}$$

where the $N \times 1$ vector r represents assets' excess returns, the $N \times K$ matrix β represents factor exposures (also referred to as factor loadings), assuming there are a total number K of factors, the $K \times 1$ vector ν is the zero-mean factor innovations, and the $N \times 1$ vector e is the zero-mean idiosyncratic errors, such that $\mathrm{E}[e|\nu] = 0$. According to the APT, the expected excess return $\mathrm{E}[r]$ can be decomposed as

$$\mathrm{E}[r] = \alpha + \beta\lambda, \tag{1.5}$$

where the $K \times 1$ vector λ is the factors' risk premia, and the $N \times 1$ vector α represents pricing errors. It is worth mentioning that Equation (1.5) indicates that the expected excess return of a given asset is determined by its exposure to a set of factors and the risk premia of those factors (plus an error term), and it explains why assets have different excess expected returns in the *cross-section*. This cross-sectional equation is both the heart and soul of empirical asset pricing and factor investing.

While theory provides a framework for studying assets' returns under (1.5), it does not offer much guidance on which factors should be included on the right-hand side of the equation. Regarding the identification of factors, there are two main approaches and both have gained a lot of attention. The first approach assumes either factors or factor exposures are known and observable. Academia and the industry take different stances in this regard. Specifically, academia considers that factors are tradable portfolios that are constructed by using some firm characteristics, such as the HML (the value factor) and SMB (the size factor) factors of Fama and French (1993) (which are built by using the book-to-market ratio and the market cap). In this case, factor realizations are the excess returns of those tradable portfolios, and factor exposures are determined by regressing assets' excess returns on factors. A related but different stance commonly adopted in the industry (e.g. the MSCI Barra model) assumes that factor exposures are observable. In this case, industry practice uses firm characteristics (usually after necessary normalization) directly as factor exposures β, and factors' risk premia can be obtained by regressing assets' return on the factor exposures.

These two distinct perspectives also clarify the different interpretations of the term "factors" as used by academia and industry, a distinction we emphasize throughout this chapter. In academia, "factors" refer to tradable portfolios. For example, the value factor, often represented as HML (High-Minus-Low), is constructed by forming two portfolios from the universe of stocks: one portfolio contains stocks with high book-to-market ratios (value stocks), and the other contains stocks with low book-to-market ratios (growth stocks). The performance of the high book-to-market portfolio minus the performance of the low book-to-market portfolio is considered the HML factor return, capturing the excess returns of value stocks over growth stocks. In contrast, within the industry, the term pertains to firm characteristics which, though not explicitly stated, serve as the factor exposures within the framework of a linear factor model. The second approach, as pioneered by Chamberlain and Rothschild (1983) and Connor and Korajczyk (1986), assumes both factors and factor exposures are latent. In this case, a statistical technique such as principal component analysis is employed to identify the factor model, and the factors are called statistical factors.

This book adopts the first approach for the following two reasons. First, the finance literature has proposed a plethora of tradable portfolios (constructed by using corresponding firm characteristics) as factors, and they have either risk-based or behavioral finance explanations. Categorizing them into different classes and providing a detailed description for each category will give the readers a comprehensive understanding of the most important and prevailing factors discovered by academia in the past several decades. Second, although the statistical factor model is